

Exploring Market Uncertainty with Google Trends

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1 Introduction

Financial markets are heavily influenced by investor sentiment and uncertainty, particularly during periods of economic stress or rapid changes in market expectations. Traditional measures of market risk often rely on price movements or volatility indices, which typically react after uncertainty has already begun affecting investor behavior. As a result, there is increasing interest in alternative data sources that provide earlier insight into shifts in market sentiment, such as Google Trends search data. Changes in search behavior reflect growing public concern surrounding economic conditions, financial stability, and market events, before these concerns are fully incorporated into asset prices. This project will explore this Google Trends data, and its applicability in identifying periods of heightened market uncertainty, and improvement to risk management decisions.

2 Data

For the data section of this project, I utilized SP 500 index data spanning from 2004 through the end of 2025. This period was selected because 2004 is the earliest point in which Google Trends data is available, ensuring consistency across all variables used in the analysis. This timeframe captures a wide range of market conditions, including periods of significant volatility and macroeconomic uncertainty, which is essential for evaluating whether Google Trends data can be used as a meaningful signal for identifying turbulent market regimes.

Alongside the SP 500 data, the only additional dataset used in this study was daily Google Trends data over the same time horizon. This parallel structure allows for a direct comparison between market movements and changes in search interest, enabling the analysis to assess the potential relationship between public attention and periods of heightened financial instability.

The training and fitting for this strategy was done on the first 12 years of

the timeframe, 2004-2016, with our out of sample testing being performed on 2016 data and onward.

3 Preprocessing

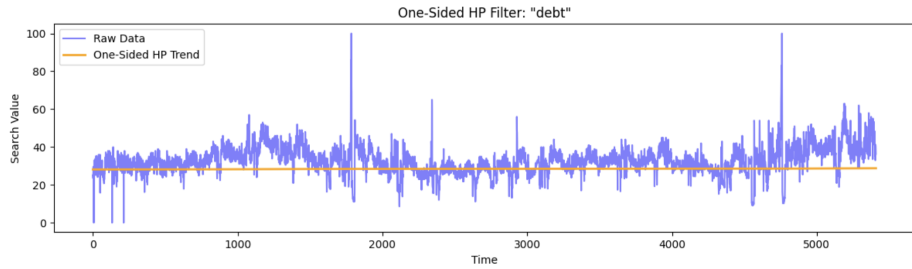
Before using Google Trends data to identify periods of market uncertainty, each search term was passed through a normalization process designed to reduce noise and isolate meaningful signals. Specifically, a simple time-series decomposition was applied of the form:

$$model = trend + seasonal$$

In this setup, the trend component captures the underlying long-term movement in search interest, while the seasonal component accounts for short-term and recurring fluctuations that are not related to broader market behavior. This step is important because it helps filter out the predictable patterns in search activity, such as sharp drops in volume around holiday seasons, most notably in late December. This ensures that these anomalies do not distort the interpretation of search behavior as a proxy for market sentiment.

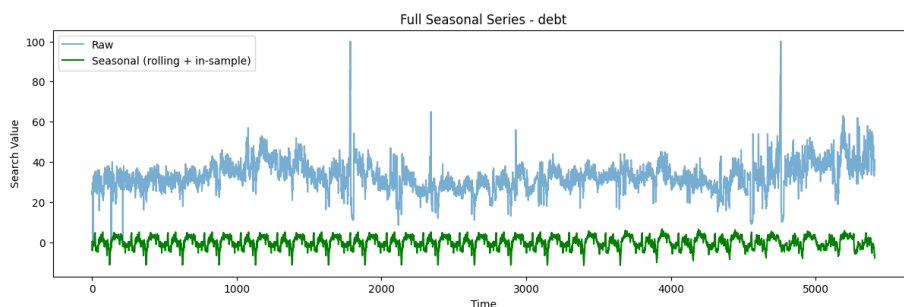
3.1 Trend

The trend line was customized to each word with the assumption that the underlying trend of the search data does not change quickly. This assumption was made because the underlying level of interest in persistent topics tends to shift slowly rather than rapidly. With this in mind, the trend component was intentionally designed to be highly smooth and to deliberately underfit the data in order to isolate long-run movement. To achieve this, a one-sided Hodrick–Prescott (HP) filter was applied with a large lambda parameter. This approach effectively dampens short-term fluctuations while preserving the underlying trend structure, and the one-sided formulation ensures that no future data is used, thereby avoiding lookahead bias.



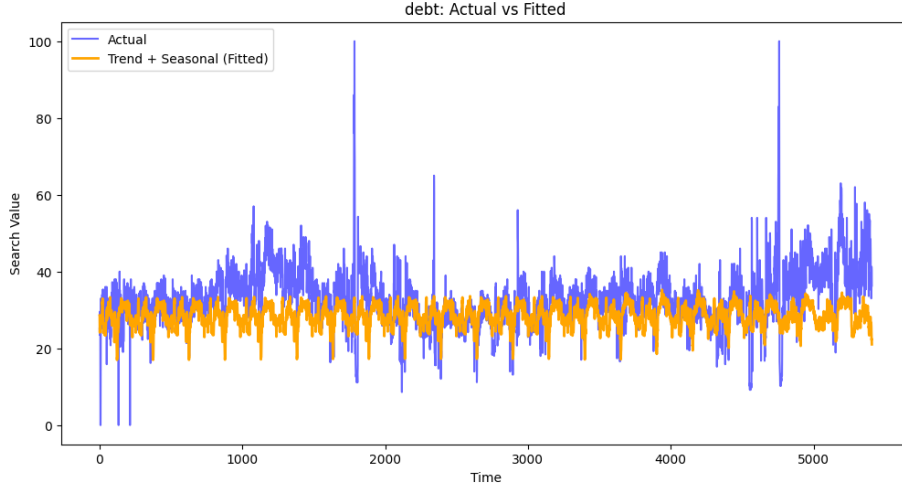
3.2 Seasonal

The seasonal component was more difficult to isolate, and was estimated using a rolling window of the previous 12 years of data as a training set. (This is the reason the first 12 years were not used in strategy testing). To identify underlying seasonal behavior, the data was first detrended by intentionally overfitting a trend component to induce stationarity. Once stationarity was achieved, the remaining structure was used to extract the dominant seasonal pattern for each search term. This process was implemented using the seasonal decompose function from the statsmodels library in Python.



3.3 Final trend data

Now that the underlying trend has been extracted as a smooth long-run component and the seasonal patterns have been identified, the two are combined to reconstruct the expected value of the series at each point in time. Subtracting this reconstructed signal from the original data yields the residual component of the model. These residuals represent the detrended and deseasonalized portion of the Google Trends data, effectively isolating short-term deviations from expected search behavior. In this sense, they serve as a normalized measure of “unexpected” attention, where typical cyclical and long-run patterns have been removed. These cleaned residuals are then used as the primary input for identifying periods of heightened market uncertainty, under the assumption that unusually large deviations in search activity reflect shifts in investor attention and perceived market risk.



4 The Detection Mechanism

Now that our trends data was filtered and normalized, I grouped the words into clusters including labor, macro, credit, oil, market, politics, and fear, with the grouping based on the significance of the word. This grouping step serves to further reduce noise by aggregating related signals, while also improving interpretability by allowing broader patterns of attention to emerge at the category level. In addition, it provides a more structured analysis of where shifts in market-related concern originate, rather than relying solely on individual search term fluctuations.

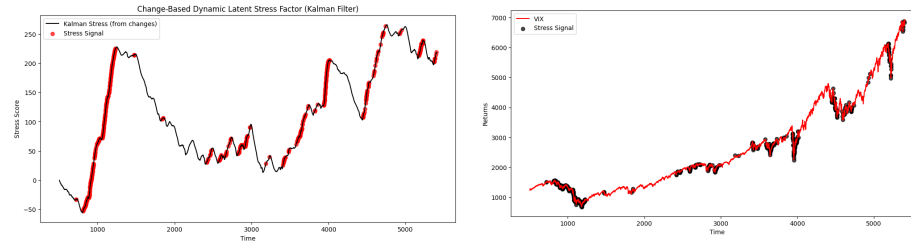
After the data was grouped, we get into the main mechanism of our detection, a Kalman filter. A Kalman filter is a recursive state-space estimation method that produces an optimal estimate of an underlying signal by combining noisy observations with a dynamically updated prediction of the system's state. It works by continuously balancing new incoming data against prior estimates, weighting each based on their relative uncertainty. This makes it particularly effective in this setting because it can smooth highly noisy, high-frequency signals like Google Trends while still remaining adaptive to changing market conditions. As a result, it provides a real-time estimate of underlying market stress without overreacting to short term spikes in search behavior. In this context, it is especially effective because, even after initial trend and seasonal filtering, residual noise remains in the data, and the Kalman filter is able to more reliably extract the underlying structure from these imperfect signals.

After implementing the Kalman filter, I initially introduced a two-part signaling mechanism to detect periods of elevated market uncertainty. The first

component is a rolling threshold based on the last 50 Kalman-filtered state estimates, which serves as a medium-term benchmark for typical market attention levels. If the current state exceeds this threshold, it suggests that search-driven attention is unusually elevated relative to recent history. The second component is a short-term z-score calculated over the past 15 days. The z-score measures how many standard deviations the current observation deviates from its recent mean, allowing the model to capture abrupt, high-frequency shifts in search activity that may not be reflected in the longer rolling threshold.

These two signals together provide complementary information, producing a strong initial indicator of market uncertainty. However, the system was overly sensitive, generating a number of false positives where the signal triggered, despite the absence of meaningful market stress. To address this, I introduced an additional filtering layer based on the first difference of the Kalman state, effectively capturing the current rate of change in attention. This measures how quickly the filtered signal is moving from one day to the next, rather than its absolute level. In testing, a threshold of 0.25 on this rate of change proved effective as a filter, as it helps distinguish genuine shifts in market attention from minor fluctuations or noise-driven oscillations.

This addition significantly improved the performance of the signal, but it still slightly struggled with some false signals, particularly in the points where people may have just been genuinely curious about a topic rather than actual uncertainty. While this additional led to significant improvement in signal performance, slight struggling with false signals persisted, likely due to searches driven by curiosity as opposed to uncertainty. I combined my uncertainty score with actual investor action through the trading volume of all SP 500 stocks (GSPC ticker). Genuine uncertainty is more likely to translate into changes in portfolio positioning, which in turn is reflected in elevated trading activity. This signal on it's own does not work very well, but when it was combined with the previous signal, it gives exceptional insight into the true uncertainty that people feel.

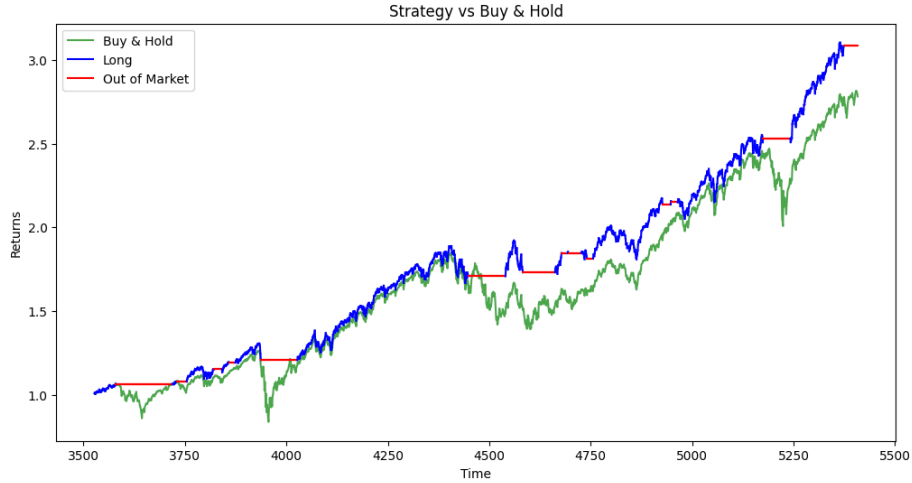


We can see that in this figure, the clusters of points come in the periods right before and during and intense drop meaning that we are now identifying

the most uncertain areas in the market.

5 Testing and Results

Now that the signal was constructed, I proceeded to test it using the SP 500 ETF over the previously defined sample period. The strategy itself is intentionally simple in order to clearly isolate the behavior and effectiveness of the risk-detection signal. The rule is straightforward, when the signal is triggered, the portfolio exits the market for a fixed holding period of 15 days; otherwise, it remains fully invested. The 15-day window is purely a design parameter and is not intended to be economically optimal in itself. It is chosen to allow sufficient time for clustered or back-to-back signals to play out without prematurely re-entering the market, while still maintaining responsiveness to new periods of detected uncertainty.



After running the backtest, the strategy slightly outperforms a basic buy-and-hold approach while remaining out of the market for nearly one-quarter of the total time. This indicates that the signal is able to reduce exposure during periods of elevated risk without sacrificing overall returns. More importantly, the strategy achieves this performance with substantially lower risk. It systematically avoids major drawdown periods, resulting in a Sharpe ratio of 1.42 compared to 0.84 for buy-and-hold. In addition, the maximum drawdown is significantly reduced, falling to 11.67 percent from 33.72 percent for the benchmark strategy. These results suggest that the model is effective at capturing periods of heightened uncertainty and translating them into meaningful downside protection.

Table 1: Overall Performance Summary

Metric	Strategy	Buy Hold	Improvement
Annualized Risk (Std)	.1108	.1966	.0858
Sharpe Ratio	1.42	.84	0.62
Max Drawdown	11.67%	33.72%	22.05%

6 Limitations

The primary limitation of this approach is that, while Google Trends data is effective at identifying periods of elevated concern, it tends to overstate their duration. In many cases, markets begin to recover before search interest normalizes, meaning the signal remains active even after the underlying risk has already subsided. As a result, the signal is best used as a tool for identifying periods of heightened uncertainty and downside risk, rather than as a complete trading system on its own. A complementary mechanism is therefore needed to more reliably determine when it is appropriate to re-enter the market.

Another limitation of this analysis is that the signal was constructed using only 11 search terms. While these words were selected to represent key dimensions of market behavior, this relatively small feature set may not fully capture the breadth of investor attention or evolving sources of uncertainty. Expanding the set of terms particularly with more nuanced or high frequency market relevant keywords could improve the robustness and sensitivity of the signal, allowing it to better reflect shifts in sentiment across a wider range of economic and financial themes.

7 Conclusion

This project demonstrates the effectiveness of analyzing human behavior in order to gain insight into the current conditions of the market. By jointly analyzing changes in search activity and corresponding market responses, the framework is able to capture periods of heightened uncertainty and translate them into meaningful risk controls. In doing so, it majorly reduces portfolio downside risk, with the maximum drawdown in testing decreasing by nearly 22 percentage points relative to a passive benchmark. Although the trading strategy implemented here is intentionally simple, the underlying detection mechanism is much more applicable. Rather than serving as a standalone trading system, this framework is best interpreted as a risk-monitoring tool that can be layered onto existing investment strategies. This signal could help then investors identify when market conditions are becoming increasingly fragile, indicating to them when a more defensive positioning should be taken.

Overall, the results suggest that attention-based signals derived from Google

Trends contain meaningful predictive structure and can serve as a valuable complement to traditional market-based indicators in managing portfolio risk.